

B.Sc. CSIT — Statistics II (STA215)

Repeated Questions: Priority Guide for Exam Preparation

Based on Past Papers: 2075, 2077, 2078, 2079, 2080, 2081 | Tribhuvan University

■ Repeated 4+ times

■ Repeated 3 times

■ Repeated 2 times

UNIT 1: Sampling Distribution & Estimation

■ Sampling Distribution of Mean / Central Limit Theorem

Appeared in: 2077, 2078, 2079, 2080, 2081 (5 times)

Key topics to master:

- Explain sampling distribution of mean with numerical example
- State and apply the Central Limit Theorem (CLT)
- Compute: population mean/variance, sample mean/variance, standard error of mean
- CLT application: probability that sample mean exceeds a value (use Z-test)

Typical Q: Population {20,21,22,23}, $n=2$, SRWR — find all sampling distributions and compare with population.

■ Sample Size Determination

Appeared in: 2075, 2077, 2078, 2079, 2080, 2081 (5+ times)

- For estimating population mean: $n = (Z \cdot \sigma / E)^2$
- For estimating population proportion: $n = Z^2 \cdot p \cdot q / E^2$
- When CV is given: use $\sigma = (CV/100) \cdot \text{mean}$

Typical Q: Estimate average with error ≤ 2 pts, $\sigma=30$, 95% confidence — find n .

■ Confidence Interval Estimation

Appeared in: 2075, 2077, 2078, 2079, 2080, 2081 (multiple)

- CI for population mean (known/unknown σ)
- CI for population proportion ($p \pm Z \cdot SE$)
- Linkage between CI and hypothesis testing

Typical Q: 400 toothpaste tubes, 20 leaked — construct 96% CI for proportion.

UNIT 2: Testing of Hypothesis

■ One-Sample Z-test / t-test for Mean

Appeared in: 2075, 2077, 2078, 2079, 2080, 2081 (every year)

- One-sample Z-test when σ known (large sample)
- One-sample t-test when σ unknown (small sample)
- State H_0 , H_1 → compute test statistic → compare with critical value → decision
- Compute p-value when asked

Typical Q: Paint drying time $\mu=122s$; sample $n=50$, $\bar{x}=116s$, $s=16.8$ — test if mean reduced. (2081)

■ Test for Proportion (Single & Two-Sample)

Appeared in: 2077, 2078, 2079, 2080, 2081 (5 times)

- Single proportion test: $Z = (p - p_0) / \sqrt{(p_0 \cdot q_0/n)}$
- Two proportion test: $Z = (p_1 - p_2) / SE$

Typical Q: 600 people, 350 have Samsung — test if equally popular. Appears in 2079 & 2080.

■ Paired Sample t-test

Appeared in: 2079, 2081 + Model Questions (3-4 times)

- Use when same subjects measured before & after
- Compute $d = x_1 - x_2$ for each pair, then $\bar{d}$ and s_d
- $t = \bar{d} / (s_d / \sqrt{n})$, $df = n-1$

Typical Q: Left vs right-hand gripping strength of 8 left-handed writers. (2081)

■ Two-Sample Independent t-test

Appeared in: 2075, 2077, 2078 (3 times)

- State assumptions clearly (normality, independence, equal/unequal variance)
- Pooled variance formula when $\sigma_1^2 = \sigma_2^2$

Typical Q: Male vs female faculty working hours, $n=60$ vs 30 . (2078)

■ Type I & Type II Errors

Appeared in: 2075, 2078, 2079 (3 times)

- Define α (Type I) = rejecting true H_0 ; β (Type II) = accepting false H_0
- Power of test = $1 - \beta$

UNIT 3: Non-Parametric Tests

■ Chi-Square Test (Independence / Association)

Appeared in: 2075, 2077, 2078, 2079, 2080, 2081 (every year)

- Set up contingency table, compute expected frequencies: $E = (\text{Row Total} \times \text{Col Total}) / \text{Grand Total}$
- $\chi^2 = \sum (O-E)^2/E$, $df = (r-1)(c-1)$
- Common contexts: email hacking vs account type, feeding vs teeth, mobile usage by age

Typical Q: Yahoo vs Gmail $\times$ Hacked/Not Hacked — test association. (2077 & 2078 — identical question!)

■ Kruskal-Wallis Test

Appeared in: 2075, 2080 (and Kruskal-Wallis/ANOVA crossover topics)

- Non-parametric alternative to one-way ANOVA
- Rank all observations, compute H statistic
- $H = [12/N(N+1)] \sum (R_j^2/n_j) - 3(N+1)$

Typical Q: Propellant burning rates for 3 missile systems. (2080)

■ Mann-Whitney U Test

Appeared in: 2077, 2078 (2 times)

- Rank combined data, compute U_1 and U_2
- Test statistic: $U = \min(U_1, U_2)$

Typical Q: IT knowledge scores for Group A vs B. Satisfaction scores Gadget A vs B.

■ Run Test for Randomness

Appeared in: 2079 (1-2 times)

- Count runs in a sequence (e.g., M/F sequence)
- Compute $Z = (r - \mu_r) / \sigma_r$, compare with critical Z

UNIT 4: Multiple Correlation & Regression

■ Multiple Linear Regression — Fitting & Interpretation

Appeared in: 2075, 2077, 2078, 2079, 2080, 2081 (every single year — Group A)

- This is the MOST important topic. Appears in Group A (10 marks) every year.
- Steps: Solve normal equations → find b_0, b_1, b_2 → write $\hat{Y} = b_0 + b_1X_1 + b_2X_2$
- Interpret b_1 : 'Holding X_2 constant, 1 unit increase in X_1 changes Y by b_1 units'
- Test significance: F-test for overall model; t-test for individual coefficients
- Compute $R^2 = SSR/SST$; Adjusted R^2 ; Standard Error of Estimate

Typical Q (2081): Job satisfaction (Y) vs aptitude (X_1) and absent days (X_2) for 10 employees.

■ ANOVA Table for Multiple Regression

Appeared in: 2075, 2079, 2080, 2081 (4 times)

- Know: $SST = SSR + SSE$; $df(\text{Regression})=k$; $df(\text{Error})=n-k-1$; $df(\text{Total})=n-1$
- $MSR = SSR/k$; $MSE = SSE/(n-k-1)$; $F = MSR/MSE$
- $R^2 = SSR/SST$; SE of estimate = $\sqrt{MSE}$

■ Partial & Multiple Correlation Coefficients

Appeared in: 2079, 2080, 2081 (3 times)

- Partial correlation $r_{12.3} = (r_{12} - r_{13} \cdot r_{23}) / \sqrt{(1-r_{13}^2)(1-r_{23}^2)}$
- Multiple correlation $R_{1.23} = \sqrt{[(r_{12}^2 + r_{13}^2 - 2 \cdot r_{12} \cdot r_{13} \cdot r_{23}) / (1-r_{23}^2)]}$

Typical Q: Given $r_{12}=0.4$, $r_{23}=0.5$, $r_{13}=0.6$ — find $R_{1.23}$ and $r_{12.3}$. (2081)

UNIT 5: Design of Experiments

■ Latin Square Design (LSD)

Appeared in: 2075, 2077, 2078, 2079, 2080, 2081 (every year)

- Conditions: p treatments → $p \times p$ layout; controls 2 nuisance factors (rows & columns)
- Why not less than 3 treatments: error $df = (p-1)(p-2)$ would be 0 or negative
- ANOVA: $SST_{\text{Total}} = SSR_{\text{Row}} + SSC_{\text{Col}} + SST_{\text{Treatment}} + SSE_{\text{Error}}$
- df : Rows= $(p-1)$, Cols= $(p-1)$, Treatments= $(p-1)$, Error= $(p-1)(p-2)$

Typical Q: 3×3 LSD layout with given cell values — perform full ANOVA. Appears almost every year!

■ One-Way ANOVA (CRD)

Appeared in: 2078, 2079, 2080, 2081 (4 times)

- H_0 : All group means are equal
- $SSB = \sum n_j(\bar{x}_{\cdot j} - \bar{x}_{\cdot \cdot})^2$; $SSW = \sum \sum (x_{ij} - \bar{x}_{\cdot j})^2$; $SST = SSB + SSW$
- $F = MSB/MSW$; compare with $F(k-1, N-k)$

Typical Q: Seminar evaluation scores by High/Mid/Low manager level. (2081)

■ Randomized Block Design (RBD)

Appeared in: 2075, 2080 (as short notes)

- Efficiency of RBD relative to CRD
- Adds blocking factor to reduce experimental error

UNIT 6: Stochastic Processes

■ Markov Chain — Definition & Characteristics

Appeared in: 2075, 2077, 2078, 2079, 2081 (5 times)

- Define Markov chain: memoryless property $P(X_{n+1}|X_0...X_n) = P(X_{n+1}|X_n)$
- Transition matrix, state space, Chapman-Kolmogorov equations
- Steady-state distribution: $\pi P = \pi$

Typical Q: Sunny/Rainy day with transition probs — forecast for Tue & Wed given Mon=Rainy. (2077)

■ Queuing Theory (M/M/1)

Appeared in: 2075, 2079, 2080, 2081 (4 times)

- Parameters: λ (arrival rate), μ (service rate), $\rho = \lambda/\mu$ (traffic intensity)
- $L_q = \rho^2/(1-\rho)$ — avg customers in queue
- $L = \rho/(1-\rho)$ — avg customers in system
- $W_q = L_q/\lambda$; $W = L/\lambda$ (Little's law)
- $P(\text{queue} > n) = \rho^{n+1}$

Typical Q: Barber shop — arrival every 12 min, service 10 min — find L_q . (2081)

■ Poisson / Binomial Counting Process

Appeared in: 2075, 2079, 2080 (3 times)

- Binomial process: $P(\text{arrival in frame}) = p$; frame length $\Delta = p/\lambda$
- Mean inter-arrival time = $1/\lambda$; Variance = $(1-p)/p^2$
- Poisson process: $P(X=k) = e^{-(\lambda)} \cdot \lambda^k / k!$

MASTER PRIORITY TABLE — All Topics Ranked

Priority	Topic	Unit	Years	Times
■ MUST	Multiple Linear Regression (fitting, interpret, test)	4	Every year	6
■ MUST	Latin Square Design (LSD) — layout & ANOVA	5	Every year	6
■ MUST	One-Sample Z/t test for mean	2	Every year	6
■ MUST	Chi-Square Test of Independence	3	Every year	6
■ MUST	Central Limit Theorem & Sampling Distribution	1	2077-2081	5
■ MUST	Sample Size Determination	1	Every year	5+
■ MUST	Markov Chain	6	2075-2081	5
■ MUST	Test for Proportion (single/two-sample)	2	2077-2081	5
■ HIGH	ANOVA Table for Regression + R^2 + F-test	4	2075,79,80,81	4

Priority	Topic	Unit	Years	Times
■ HIGH	One-Way ANOVA (CRD)	5	2078-2081	4
■ HIGH	Queuing Theory (M/M/1 system)	6	2075,79,80,81	4
■ HIGH	Partial & Multiple Correlation	4	2079-2081	3
■ HIGH	Paired Sample t-test	2	2079,2081	3
■ MED	Kruskal-Wallis Test	3	2075,2080	2
■ MED	Two-Sample Independent t-test	2	2075,77,78	3
■ MED	Confidence Interval (mean & proportion)	1	Multiple	4
■ MED	Mann-Whitney U Test	3	2077,2078	2
■ MED	Poisson / Binomial Counting Process	6	2075,79,80	3
■ LOW	Run Test for Randomness	3	2079	1
■ LOW	Randomized Block Design (RBD)	5	2075,2080	2
■ LOW	Type I & Type II Errors (definitions)	2	2075,78,79	3

Exam Strategy Tips:

- Group A (3 Qs, attempt 2×10 marks): Always has Multiple Regression, LSD, and Hypothesis Testing.
- Group B (8 Qs $\times$ 5 marks): Covers all units. Prioritize ■ topics — they repeat every 1-2 years.
- Identical question appeared in 2077 & 2078: Yahoo/Gmail Chi-square — memorize the solution steps.
- For M/M/1 queuing: always check $\rho = \lambda/\mu$ first. Must be < 1 for stable system.
- Markov chain: practice both numerical transition matrix problems and written definition questions.